CarDekho Used Car Price Analysis

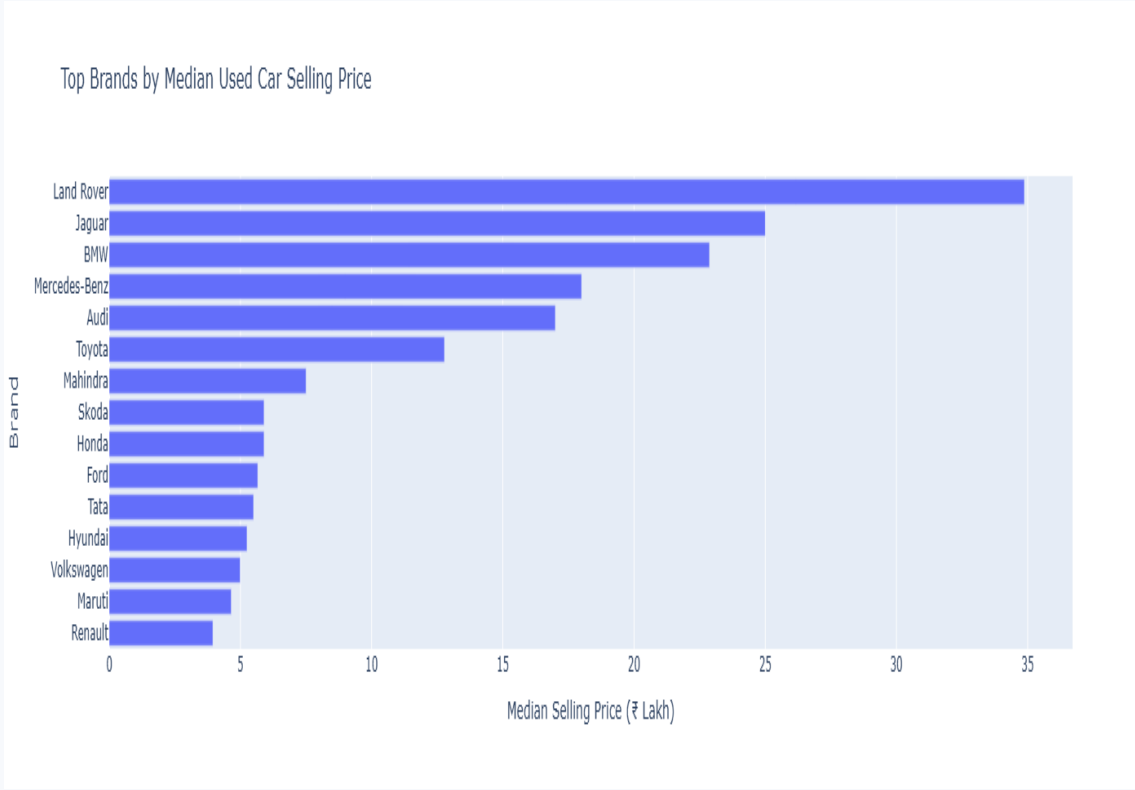
Depreciation, value retention and regression modeling for used-car pricing intelligence

Python | Plotly | Scikit-learn | Power BI

15K+
Listings
used cars

0.8086
Best R^2
Gradient Boosting

₹4.80L
Best RMSE
Gradient Boosting



Business question

What drives resale prices, how does value change with age, and how accurately can listing prices be estimated?

Business Problem & Objectives

Turning used-car listings into pricing intelligence

Marketplace challenge

Pricing used vehicles is difficult because age, mileage, brand, performance, fuel, transmission and market segment interact nonlinearly.

- Identify major resale-price drivers
- Measure cross-sectional market depreciation
- Compare brand-level value retention
- Build and compare regression models
- Translate findings into Power BI

Analytical outcome

Gradient Boosting

Best overall model

highest R^2 + lowest RMSE

₹1.16L

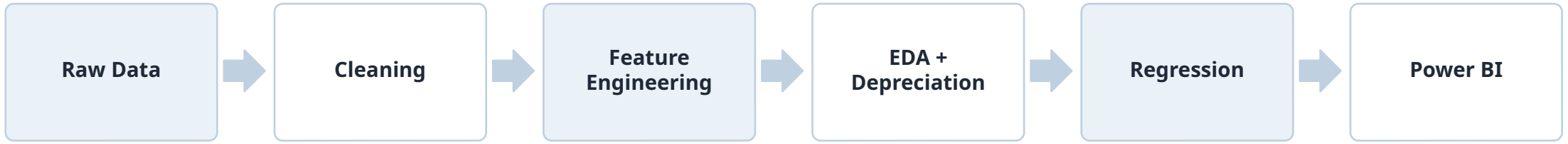
Lowest MAE

Random Forest

The final dashboard connects market context, depreciation, and model performance so the analysis reads as a business story rather than a standalone ML exercise.

Data Preparation & Workflow

A reproducible path from raw listings to dashboard-ready outputs



2
Duplicates removed
plus index column

2
Invalid seat records fixed
Honda City + Nissan Kicks

300,000
KM business cap
extreme outliers excluded

5+
Engineered features
age/km/value metrics

Core variables: brand, model, age, kilometres driven, fuel, transmission, mileage, engine, max power, seats and selling price.

What Drives Used-Car Prices?

Numerical relationships and segment effects

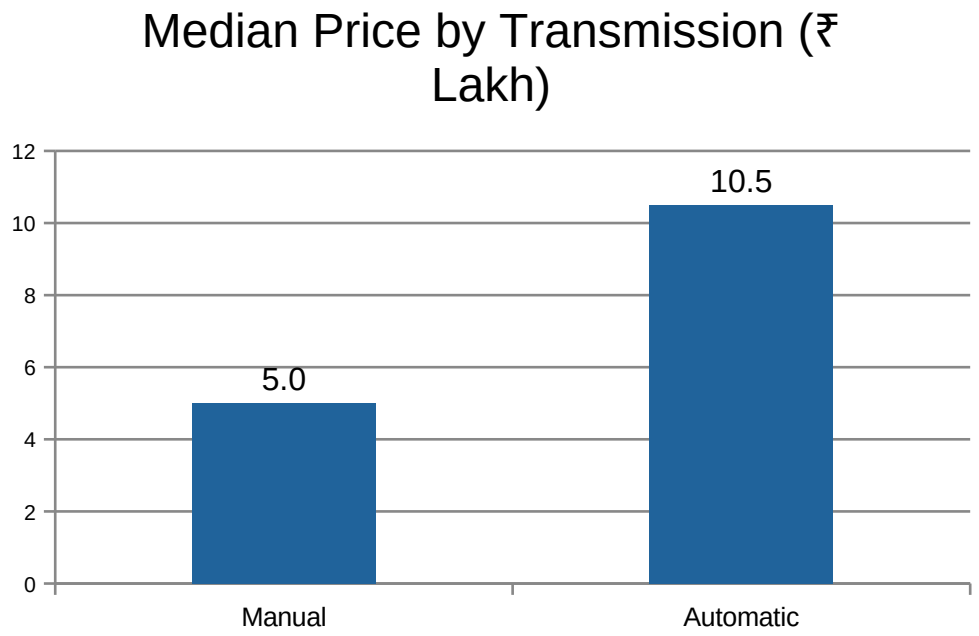
+0.751
Max Power vs Price
strongest numerical correlate

+0.587
Engine vs Price
strong positive

-0.306
Mileage vs Price
negative association

-0.241
Age vs Price
stronger than km alone

-0.121
KM vs Price
weak negative



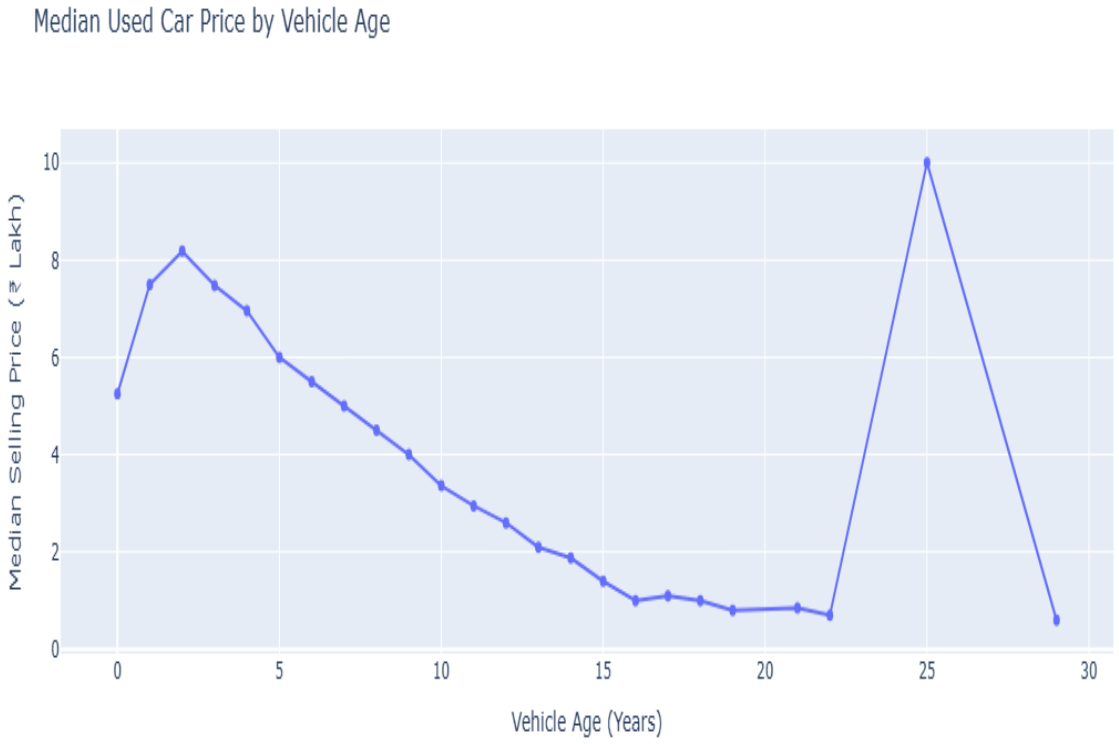
Raw automatic premium

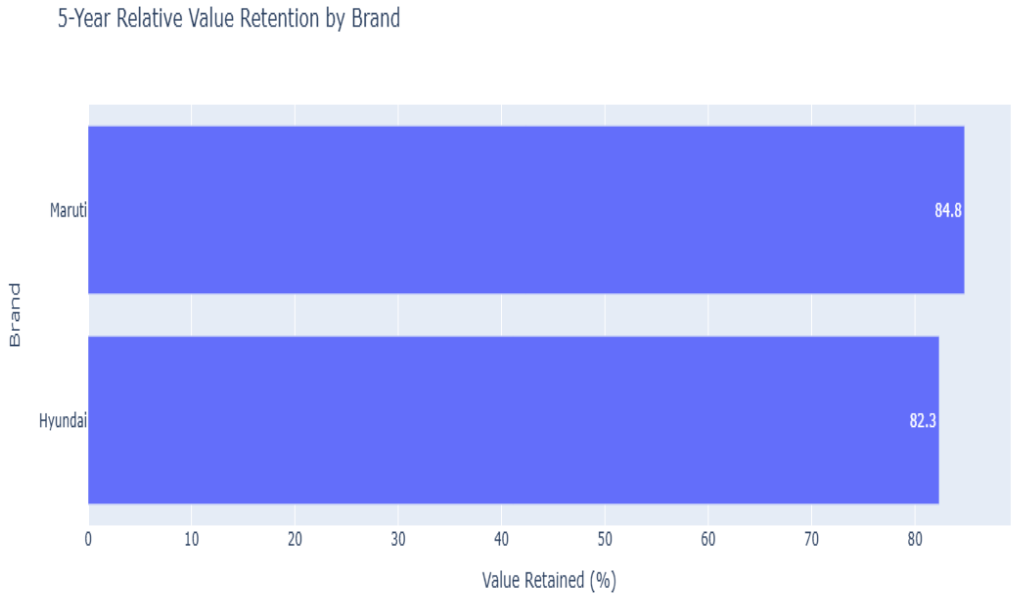
110%

Interpret with caution: automatic cars may also be newer, more powerful, or concentrated in premium brands. The regression model helps control for multiple factors simultaneously.

Depreciation & Value Retention

Cross-sectional market depreciation, not original-MSRP depreciation





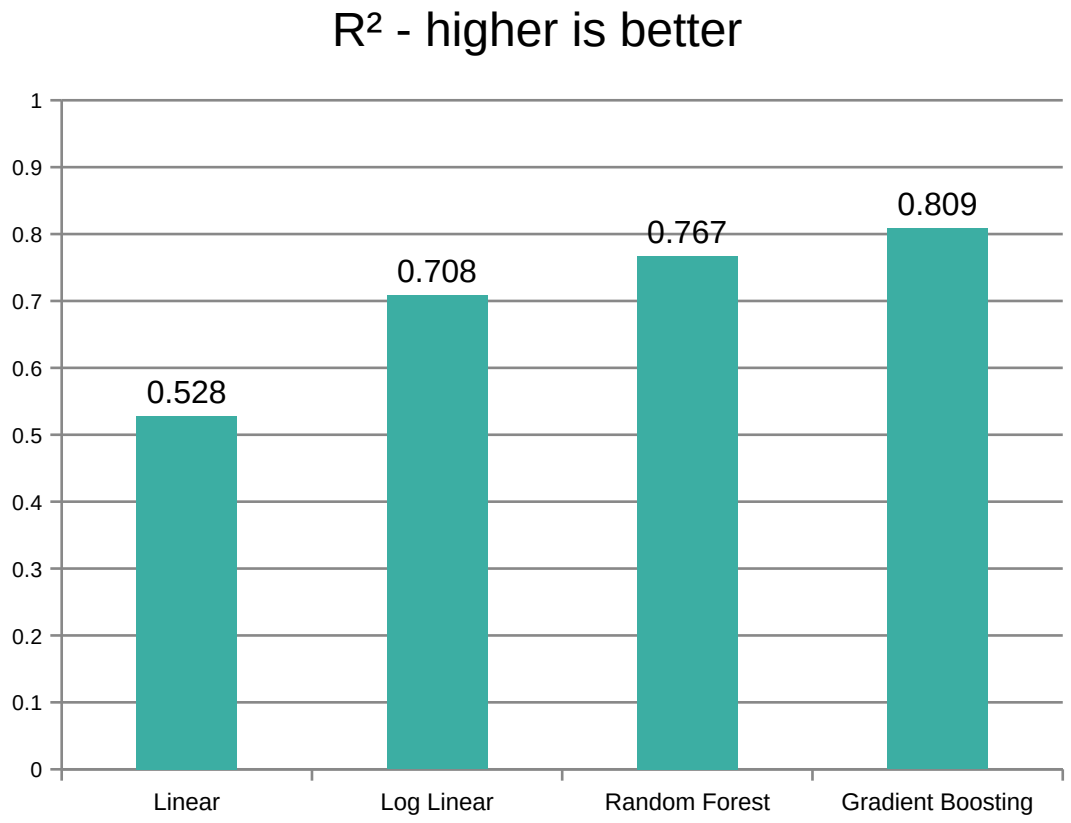
5-year relative value retention

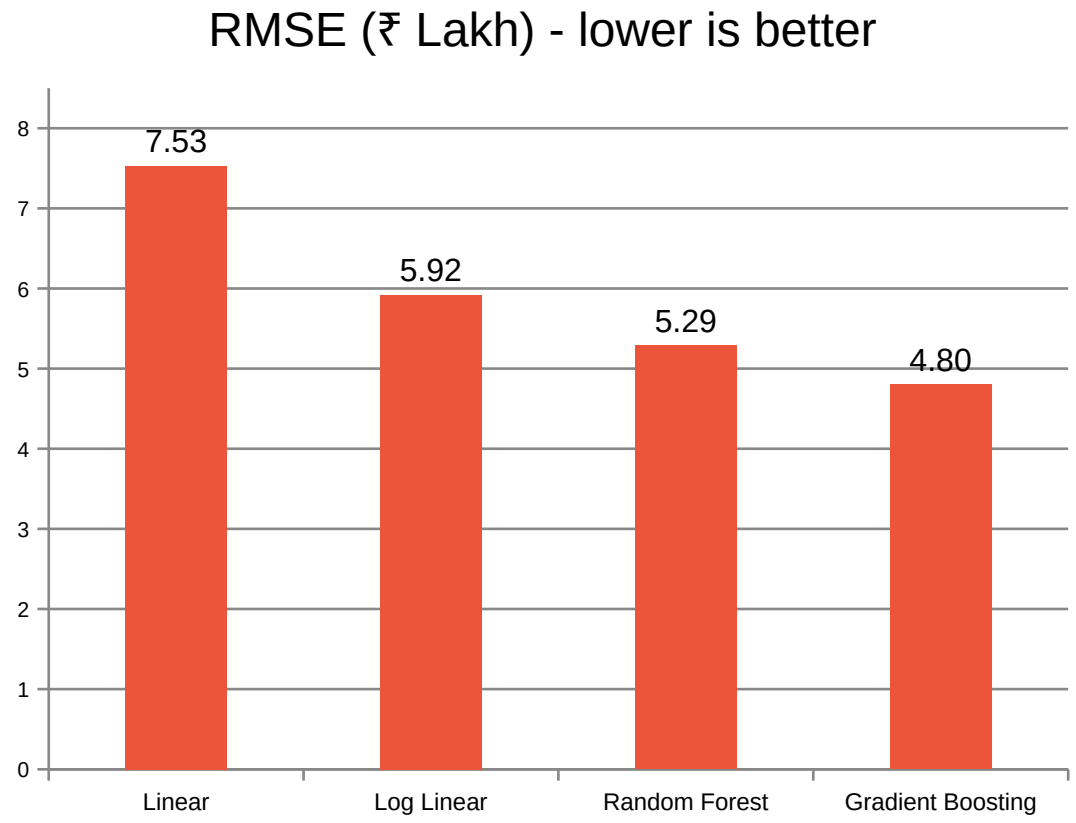
Maruti 84.8% | Hyundai 82.3%

Older age groups with very small samples were filtered to avoid misleading spikes.

Regression Model Comparison

Nonlinear methods materially improved predictive performance





Gradient Boosting
Final model
best overall

0.8086
Best R²
Gradient Boosting

₹4.80L
Best RMSE
Gradient Boosting

₹1.16L
Best MAE
Random Forest

Power BI Dashboard Story

Three pages designed for executive exploration

PAGE 1

Market Overview

KPIs, brand listings, median price, fuel mix, transmission mix, price bands

Marketplace context

PAGE 2

Depreciation & Value Retention

Price by vehicle age, age-group pricing, five-year retention, year-on-year market depreciation

Depreciation story

PAGE 3

Regression & Pricing Intelligence

Model KPIs, performance comparison, feature importance, actual vs predicted, error by brand

Model intelligence

Recommendations & Final Takeaway

How this analysis can support used-car marketplace decisions

Use vehicle age, maximum power, engine capacity, brand and transmission as core pricing inputs.

Interpret kilometres driven together with vehicle age rather than independently.

Prefer nonlinear models for used-car price estimation.

Monitor error by brand and consider specialized models for weak segments.

Use depreciation analysis to improve seller guidance and buyer transparency.

FINAL TAKEAWAY

Gradient Boosting explained about 81% of observed price variation.

The project combines market analysis, depreciation, machine learning and Power BI into one end-to-end pricing intelligence workflow.

Limitations: listing price may differ from transaction price; original MSRP and detailed vehicle condition/history are unavailable.